LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	42
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Explanation of helicopter autopilot system and various facilities available through it.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Explanation of helicopter autopilot system and various facilities available through it.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS block diagram and facility list from IB; Instructor supervision during Q&A.;
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an autopilot system?
(b) Name two manual flight controls used in a helicopter.
(c) Why is an autopilot necessary for helicopter operations?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and basic principle of AFCS • Sensors and data acquisition units • Control laws and stability augmentation • Actuator types and power sources • System integration with flight instruments	L & Ds	PPT, IB	15 min
(b)	• Attitude hold and heading hold functions • Altitude hold and vertical speed control • Yaw damping and turn coordination • Auto-hover and auto-landing facilities • Manual override and disengagement procedures	L & Ds	PPT, PDF	20 min
(c)	• System limitations and failure modes • Maintenance requirements and checks • Operational restrictions during flight • Troubleshooting common AFCS faults • Future trends in helicopter automation	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the AFCS maintains altitude during flight.
(b) Describe the procedure for manually overriding the autopilot.
(c) List three limitations of the current helicopter autopilot system.

12. Home Assignment and Follow Up	Prepare a brief note on the need for Flight Director Systems (FDS) as covered in the next lesson plan (LP 43), highlighting how it complements the AFCS.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Need and components of FDS.	Refer to the master syllabus reference list	LP 43

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Captain, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min